

Expectation, Laws, Tail Integrals, and Finite Conditioning

Self-contained offline study packet – 17–19 July 2026

This packet deploys three interfaces which cooperate but must not be collapsed into one causal chain:

$$Y \in L^1(\mathbb{P}) \implies \mathbb{E}[Y] \text{ is finite and } Y\mathbb{P} \text{ is finite signed,}$$
$$X \mapsto \mu_X = X_{\#}\mathbb{P} \implies \mathbb{E}[\varphi(X)] = \int \varphi d\mu_X,$$

$\mathbb{P}_B$ and finite partitions $\implies$ Bayes' formula and finite-partition conditional expectation.

The first line is an admissibility gate, the second is a pushforward representation, and the third is normalized restriction. A law exists without a Lebesgue density, and expectation is not defined through Radon–Nikodym.

Contract

- **Objects:** expectation as an integral, pushforward laws, transformed payments, and finite conditioning.
- **Purpose:** deploy the L^1 /density interface into probability calculations while retaining the exact measure-theoretic boundary.
- **Regime:** ENS L3+/early M1 theorem discipline with Exam P mechanisms used only as terminal checks.
- **Required output:** three closed-book attempts, a representation ledger, the first point of rupture, and a minimal repair.
- **Boundary:** the full Hahn–first proof of Radon–Nikodym and general conditional densities are deferred.

Session 1 – Friday, 17 July, 90 minutes

1. 15 minutes: complete Exercise 0 without notes.
2. 25 minutes: read Lecture Notes, Sections 1–3.
3. 40 minutes: complete Exercise 1.
4. 10 minutes: enter the law $\mu_X = X_{\#}\mathbb{P}$ in the representation ledger, and add a pdf row only under an explicit domination hypothesis.

Required output. Reconstruct LOTUS from the pushforward definition; distinguish nonnegative expectation, defined signed expectation, and finite L^1 expectation; explain why a law needs no density.

Session 2 – Saturday, 18 July, 120 minutes

1. 20 minutes: read Lecture Notes, Sections 4–5.
2. 25 minutes: prove the subgraph measurability and Tonelli identity in Exercise 2(a)–(b).
3. 35 minutes: complete the transformed-payment parts of Exercise 2.
4. 25 minutes: complete Exercise 5(a)–(b), the capped-payment and Poisson deployment analogues.
5. 15 minutes: reconstruct the cap, deductible, and limited-loss tail identities without notes.

Required output. Identify $g(X)$ before taking its expectation; use Jensen only as a comparison theorem, never as a substitute for the expectation calculation.

Session 3 – Sunday, 19 July, 120 minutes

1. 25 minutes: read Lecture Notes, Sections 6–8.
2. 55 minutes: complete Exercises 3–4.
3. 20 minutes: complete Exercise 5(c), the finite-mixture deployment analogue.
4. 20 minutes: complete the work record. Open the independent solutions only after a complete attempt.

Required output. Derive Bayes' formula from normalized joint masses and derive total expectation from finite-partition conditional expectation.

Attempt discipline

Use `main.tex` until every assigned exercise has a complete attempt. The independent solution document is compiled from `solutions_main.tex`. When a rupture occurs, repair the first step that prevents a legitimate continuation; do not restart the entire theory.

Self-contained lecture notes

Admissibility, representation, transformation, and finite conditioning

These notes contain every definition, theorem statement, and proof interface required by the exercises. Self-contained does not mean encyclopedic: the signed Radon–Nikodym theorem is stated in the exact finite regime in which it is used, while its Hahn–Jordan construction remains a declared deferred proof debt.

1. The probability-space L^p gate

Let $(E, \mathcal{A}, \mu)$ be a finite measure space. If $1 \leq p < q < \infty$, Hölder's inequality with exponents q/p and $q/(q-p)$ gives

$$\|f\|_p^p = \int_E |f|^p \mathbf{1}_E d\mu \leq \left(\int_E |f|^q d\mu \right)^{p/q} \mu(E)^{1-p/q}.$$

Hence

$$\boxed{\|f\|_p \leq \mu(E)^{1/p-1/q} \|f\|_q}, \quad L^q(\mu) \subseteq L^p(\mu).$$

For $q = \infty$, the same conclusion follows from

$$\int_E |f|^p d\mu \leq \|f\|_\infty^p \mu(E).$$

On a probability space this becomes

$$L^\infty(\mathbb{P}) \subseteq L^q(\mathbb{P}) \subseteq L^p(\mathbb{P}) \subseteq L^1(\mathbb{P}), \quad \|f\|_p \leq \|f\|_q \quad (p < q).$$

There is no corresponding inclusion on an arbitrary infinite measure space.

Let $Y \in L^1(\mathbb{P})$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -field. Define on $(\Omega, \mathcal{G})$

$$\nu_Y^\mathcal{G}(A) = \mathbb{E}[Y \mathbf{1}_A], \quad A \in \mathcal{G}.$$

Writing $Y = Y^+ - Y^-$ shows that $\nu_Y^\mathcal{G}$ is a finite signed measure. Moreover,

$$|\nu_Y^\mathcal{G}|(\Omega) \leq \mathbb{E}[|Y|] < \infty, \quad \nu_Y^\mathcal{G} \ll \mathbb{P}|_\mathcal{G}.$$

Signed Radon–Nikodym interface. If μ is σ -finite and ν is a finite signed measure with $\nu \ll \mu$, then there exists $h \in L^1(\mu)$ such that

$$\nu(A) = \int_A h d\mu \quad \text{for every measurable } A,$$

and h is unique μ -almost everywhere. Applied on $(\Omega, \mathcal{G})$, this gives a $\mathcal{G}$ -measurable $H \in L^1(\mathbb{P}|_\mathcal{G})$, unique $\mathbb{P}$ -almost surely, such that

$$\mathbb{E}[Y \mathbf{1}_A] = \int_A H d\mathbb{P}, \quad A \in \mathcal{G}.$$

This representative is $\mathbb{E}[Y | \mathcal{G}]$. The numerator, reference measure, measurable space, and almost-everywhere basis are all part of the object.

2. Expectation and its domain of definition

For a measurable $X : \Omega \rightarrow \overline{\mathbb{R}}$, write

$$X^+ = \max\{X, 0\}, \quad X^- = \max\{-X, 0\}.$$

The regimes are distinct:

- If $X \geq 0$, then $\mathbb{E}[X] \in [0, +\infty]$ is always defined.
- For signed X , the expression $\mathbb{E}[X^+] - \mathbb{E}[X^-]$ is defined unless both terms are $+\infty$.
- If $X \in L^1(\mathbb{P})$, then $\mathbb{E}[X]$ is finite and $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$.

Thus “defined” and “finite” are not synonyms. A nonnegative random variable may have expectation $+\infty$; a signed random variable may have no expectation at all because $+\infty - (+\infty)$ is not defined.

3. Laws and the pushforward integration formula

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (S, \mathcal{S})$ be measurable. Its law is the probability measure

$$\mu_X = X_\# \mathbb{P}, \quad \mu_X(C) = \mathbb{P}(X \in C), \quad C \in \mathcal{S}.$$

LOTUS. For every measurable $\varphi : S \rightarrow [0, +\infty]$,

$$\boxed{\mathbb{E}[\varphi(X)] = \int_S \varphi d\mu_X}.$$

The same identity holds for signed φ whenever the integral is defined.

Proof. For $C \in \mathcal{S}$,

$$\mathbb{E}[\mathbf{1}_C(X)] = \mathbb{P}(X \in C) = \mu_X(C) = \int_S \mathbf{1}_C d\mu_X.$$

Linearity gives the identity for nonnegative simple functions. If $\varphi \geq 0$ is measurable, choose nonnegative simple functions $\varphi_n \uparrow \varphi$. Then $\varphi_n(X) \uparrow \varphi(X)$, so monotone convergence on both spaces yields

$$\mathbb{E}[\varphi(X)] = \lim_n \mathbb{E}[\varphi_n(X)] = \lim_n \int_S \varphi_n d\mu_X = \int_S \varphi d\mu_X.$$

For signed φ , apply the nonnegative result to φ^+ and φ^- whenever their difference is defined. $\square$

If μ_X is atomic, the integral is a weighted sum. If $\mu_X \ll \lambda$ on $\mathbb{R}^d$, Radon–Nikodym supplies a density $f_X = d\mu_X/d\lambda$, unique λ -almost everywhere, and

$$\mathbb{E}[\varphi(X)] = \int_{\mathbb{R}^d} \varphi(x) f_X(x) dx.$$

The density formula is a representation of LOTUS under an additional domination hypothesis. The law itself needs no chosen reference measure and no density.

4. Tonelli and the layer-cake representation

Let $(E, \mathcal{A}, \mu)$ and $(F, \mathcal{B}, \nu)$ be σ -finite measure spaces. If $f : E \times F \rightarrow [0, +\infty]$ is measurable, Tonelli's theorem gives

$$\int_{E \times F} f d(\mu \otimes \nu) = \int_E \left(\int_F f(x, y) \nu(dy) \right) \mu(dx) = \int_F \left(\int_E f(x, y) \mu(dx) \right) \nu(dy),$$

with all three values allowed to be $+\infty$. Positivity is decisive. For a signed integrand, Fubini requires absolute integrability; otherwise an exchange may conceal the undefined expression $+\infty - (+\infty)$.

Let $f : E \rightarrow [0, +\infty]$ be measurable and set

$$H_f = \{(x, t) \in E \times (0, +\infty) : t < f(x)\}.$$

The subgraph is measurable because

$$H_f = \bigcup_{q \in \mathbb{Q}_{>0}} \{x : f(x) > q\} \times (0, q).$$

Since

$$f(x) = \int_0^\infty \mathbf{1}_{\{t < f(x)\}} dt,$$

Tonelli applied to $\mathbf{1}_{H_f}$ gives

$$\boxed{\int_E f d\mu = \int_0^\infty \mu(f > t) dt}.$$

More generally, for $p > 0$,

$$\int_E f^p d\mu = p \int_0^\infty t^{p-1} \mu(f > t) dt.$$

On a probability space, for $X \geq 0$,

$$\boxed{\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) dt}.$$

The identity remains valid when the common value is $+\infty$.

5. Transformed insurance payments

For $X \geq 0$, a deductible $d > 0$, and a cap $u > 0$, apply the tail formula to the transformed variable, not to its mean. Since

$$\mathbb{P}((X - d)_+ > t) = \mathbb{P}(X > d + t),$$

one obtains

$$\boxed{\mathbb{E}[(X - d)_+] = \int_d^\infty \mathbb{P}(X > s) ds}.$$

Similarly,

$$\boxed{\mathbb{E}[X \wedge u] = \int_0^u \mathbb{P}(X > s) ds},$$

and the limited-loss payment satisfies

$$\mathbb{E}[\min\{(X - d)_+, u\}] = \int_d^{d+u} \mathbb{P}(X > s) ds.$$

If $X \sim \text{Exp}(\lambda)$, these become

$$\mathbb{E}[(X - d)_+] = \frac{e^{-\lambda d}}{\lambda}, \quad \mathbb{E}[X \wedge u] = \frac{1 - e^{-\lambda u}}{\lambda},$$

and

$$\mathbb{E}[\min\{(X - d)_+, u\}] = \frac{e^{-\lambda d}(1 - e^{-\lambda u})}{\lambda}.$$

If g is convex, $X \in L^1$, and $g(X) \in L^1$, Jensen's inequality gives

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

It gives a comparison, not the value of $\mathbb{E}[g(X)]$.

6. Conditioning on a positive-probability event

For $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, define the normalized restriction

$$\mathbb{P}_B(A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad A \in \mathcal{F}.$$

Then $\mathbb{P}_B \ll \mathbb{P}$ and

$$\frac{d\mathbb{P}_B}{d\mathbb{P}} = \frac{\mathbf{1}_B}{\mathbb{P}(B)} \quad \mathbb{P}\text{-almost surely.}$$

For $Y \in L^1(\mathbb{P})$,

$$\mathbb{E}[Y | B] = \frac{\mathbb{E}[Y \mathbf{1}_B]}{\mathbb{P}(B)}.$$

The ratio formula does not condition on a null event. If X has a continuous law, then typically $\mathbb{P}(X = x) = 0$; conditioning on $X = x$ requires a conditional kernel or density and a version choice, which lies beyond this packet.

7. Finite partitions, total expectation, and Bayes

Let $(B_i)_{1 \leq i \leq r}$ be a measurable partition and set $\mathcal{G} = \sigma(B_1, \dots, B_r)$. Every finite real-valued $\mathcal{G}$ -measurable random variable is constant on each atom, hence has the form

$$H = \sum_{i=1}^r c_i \mathbf{1}_{B_i}.$$

For $Y \in L^1(\mathbb{P})$, the conditional-expectation identities

$$\int_A H d\mathbb{P} = \int_A Y d\mathbb{P}, \quad A \in \mathcal{G},$$

determine the coefficients on positive-probability atoms:

$$\mathbb{E}[Y | \mathcal{G}] = \sum_{i: \mathbb{P}(B_i) > 0} \frac{\mathbb{E}[Y \mathbf{1}_{B_i}]}{\mathbb{P}(B_i)} \mathbf{1}_{B_i}.$$

On a null atom the coefficient is arbitrary; conditional expectation is an almost-sure equivalence class.

Taking expectations gives the law of total expectation,

$$\mathbb{E}[Y] = \sum_i \mathbb{P}(B_i) \mathbb{E}[Y | B_i],$$

where null-atom terms contribute zero. Setting $Y = \mathbf{1}_A$ gives total probability,

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

If $\mathbb{P}(A) > 0$, normalization of the joint masses gives Bayes' formula:

$$\mathbb{P}(B_j | A) = \frac{\mathbb{P}(A | B_j) \mathbb{P}(B_j)}{\sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i)}.$$

Bayes and total expectation are consequences of the same finite-partition structure, not additional axioms.

8. Representation ledger

A formula is not enough to identify a density. Record its numerator, reference measure, domain, and uniqueness basis. A pushforward law is a measure in its own right and therefore occupies a representation row, not a malformed density row.

Object	Numerator or construction	Reference	Domain	Equality basis
$d(f\mathbb{P})/d\mathbb{P} = f$	$A \mapsto \mathbb{E}[f\mathbf{1}_A]$	$\mathbb{P}$	$(\Omega, \mathcal{F})$	$\mathbb{P}$ -a.s.
$\mu_X = X_{\#}\mathbb{P}$	pushforward of $\mathbb{P}$ by X	none	$(S, \mathcal{S})$	equality of measures
$f_X = d\mu_X/d\lambda$	law μ_X , only if $\mu_X \ll \lambda$	λ	state space	λ -a.e.
$d\mathbb{P}_A/d\mathbb{P}$	$C \mapsto \mathbb{P}(C \cap A)/\mathbb{P}(A)$	$\mathbb{P}$	$(\Omega, \mathcal{F})$	$\mathbb{P}$ -a.s.
$\mathbb{E}[Y \mid \mathcal{G}]$	$C \mapsto \mathbb{E}[Y\mathbf{1}_C], C \in \mathcal{G}$	$\mathbb{P} _{\mathcal{G}}$	$(\Omega, \mathcal{G})$	$\mathbb{P}$ -a.s.

The same algebraic expression may represent different objects on different measurable spaces. The ambient measure and almost-everywhere relation are not decorative annotations; they are part of the statement.

Exercise sheet

Complete every exercise before opening the independent solutions

Exercise 0 – The L^p –RN retention gate

Let $q > 1$, let $Y \in L^q(\mathbb{P})$, and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -field. Define

$$\nu_Y^{\mathcal{G}}(A) = \mathbb{E}[Y\mathbf{1}_A], \quad A \in \mathcal{G}.$$

1. State why $Y \in L^1(\mathbb{P})$ and give the exact norm inequality.
2. Identify the numerator measure, reference measure, and measurable space. State why the numerator is finite signed and absolutely continuous with respect to the reference measure; do not re-prove countable additivity.
3. State the signed Radon–Nikodym conclusion, including the space containing the representative and its uniqueness relation.
4. Let X be a random element with law $\mu_X = X_{\#}\mathbb{P}$. Explain why LOTUS requires no assumption $\mu_X \ll \lambda$, whereas a density representation $f_X = d\mu_X/d\lambda$ does.

Limit: 15 minutes. This is a recognition gate; the full Hahn–first construction is not requested.

Exercise 1 – Expectation through the law

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (S, \mathcal{S})$ be measurable and let $\mu_X = X_{\#}\mathbb{P}$.

1. Prove for indicators, then for nonnegative simple functions, that

$$\mathbb{E}[\varphi(X)] = \int_S \varphi d\mu_X.$$

2. Extend the formula to nonnegative measurable φ and state the convergence theorem used.
3. Extend it to signed φ whenever the integral is defined.
4. Give one example of a nonnegative X with $\mathbb{E}[X] = +\infty$, and one signed X for which $\mathbb{E}[X]$ is undefined because both parts have infinite expectation.
5. Explain why discrete sums and continuous density integrals are representations of this identity rather than definitions of different expectations.

Exercise 2 – Subgraphs, tail integration, and insurance payments

Let $(E, \mathcal{A}, \mu)$ be a σ -finite measure space and let $f : E \rightarrow [0, +\infty]$ be measurable.

1. Prove that

$$H_f = \{(x, t) \in E \times (0, +\infty) : t < f(x)\}$$

is $\mathcal{A} \otimes \mathcal{B}((0, +\infty))$ -measurable.

2. Apply Tonelli to $\mathbf{1}_{H_f}$ and prove

$$\int_E f d\mu = \int_0^\infty \mu(f > t) dt.$$

Now let $X \geq 0$ be a random variable and let $d, u > 0$.

3. Prove

$$\mathbb{E}[(X - d)_+] = \int_d^\infty \mathbb{P}(X > t) dt, \quad \mathbb{E}[X \wedge u] = \int_0^u \mathbb{P}(X > t) dt.$$

4. If $X \sim \text{Exp}(\lambda)$, compute both expectations.

5. Determine $\mathbb{E}[\min\{(X - d)_+, u\}]$ in tail form and then for an exponential loss.

6. State one convexity regime in which Jensen compares $\mathbb{E}[g(X)]$ with $g(\mathbb{E}[X])$, and explain why the comparison does not compute the expected payment.

Exercise 3 – Conditioning on a finite partition

Let $(B_i)_{1 \leq i \leq r}$ be a measurable partition and let $Y \in L^1(\mathbb{P})$. Assume first that $\mathbb{P}(B_i) > 0$ for every i .

1. Determine the general form of a $\sigma(B_1, \dots, B_r)$ -measurable finite real-valued random variable.

2. Use the conditional-expectation integral identities to prove

$$\mathbb{E}[Y \mid \sigma(B_1, \dots, B_r)] = \sum_i \frac{\mathbb{E}[Y \mathbf{1}_{B_i}]}{\mathbb{P}(B_i)} \mathbf{1}_{B_i}.$$

3. Deduce the law of total expectation.

4. Treat a partition containing a null atom and identify exactly what remains undetermined.

5. For an event A , apply the formula to $Y = \mathbf{1}_A$ and recover the law of total probability.

Exercise 4 – Bayes and representation bookkeeping

Let $(B_i)_{1 \leq i \leq r}$ be a partition with positive probabilities and let A satisfy $\mathbb{P}(A) > 0$.

1. Derive Bayes' formula from the joint masses $\mathbb{P}(A \cap B_i)$.

2. Define $\mathbb{P}_A(C) = \mathbb{P}(C \cap A)/\mathbb{P}(A)$ and prove

$$\frac{d\mathbb{P}_A}{d\mathbb{P}} = \frac{\mathbf{1}_A}{\mathbb{P}(A)}.$$

3. For $Y \in L^1(\mathbb{P})$ and $\mathcal{G} = \sigma(B_1, \dots, B_r)$, fill in the numerator measure, reference measure, domain, and almost-everywhere basis for

$$\frac{d\mathbb{P}_A}{d\mathbb{P}}, \quad \mathbb{E}[Y \mid \mathcal{G}].$$

4. Explain why $\mathbb{P}(C \mid X = x)$ cannot in general be defined by an event ratio when X has a continuous law.

Exercise 5 – Original deployment analogues

The following are original problems. The labels record only the structural mechanism of the corresponding official-question anchors; no official problem statement is reproduced.

1. **Capped-payment mechanism (Q50 anchor).** A nonnegative loss X satisfies

$$\mathbb{P}(X > t) = (1 + t)^{-2}, \quad t \geq 0.$$

Identify the transformed payment and compute $\mathbb{E}[X \wedge 1]$.

2. **Nonlinear Poisson moment (Q386 anchor).** Let $N \sim \text{Poisson}(2)$ and let the payment be

$$g(N) = \frac{N(N+1)}{2}.$$

Compute $\mathbb{E}[g(N)]$ and state why $g(\mathbb{E}[N])$ is not the required object.

3. **Finite-mixture inversion (Q370 anchor).** A policy belongs to class B_1 with probability 0.3 and to class B_2 with probability 0.7. An observed event A satisfies

$$\mathbb{P}(A \mid B_1) = 0.8, \quad \mathbb{P}(A \mid B_2) = 0.2.$$

Form the two joint masses and compute $\mathbb{P}(B_1 \mid A)$.

Closed-book reconstruction

Without notes, complete the two chains

$$\boxed{X} \mapsto \boxed{\mu_X} \mapsto \boxed{\int \varphi d\mu_X}$$

and



For every density encountered, name the numerator, reference measure, domain, and almost-everywhere basis. For μ_X , explain why those density fields are not applicable until a reference measure is chosen.

Deferred proof debts

The full Hahn-first construction of Radon–Nikodym, σ -finite gluing, and general conditional kernels or conditional densities are not completion requirements for this packet.

Work and rupture record

Record the first illegitimate step, not a general feeling of difficulty

Session record

Session	Actual time	State	First rupture and minimal repair
17 July		not attempted	
18 July		not attempted	
19 July		not attempted	

Exercise 0 – Retention gate

$L^q(\mathbb{P}) \subseteq L^1(\mathbb{P})$ and exact inequality:

Numerator, reference measure, and measurable space:

Signed RN conclusion and almost-sure relation:

Why LOTUS needs no density:

First interface that could not be recovered:

Exercise 1 – Expectation and law

Indicators $\rightarrow$ simple $\rightarrow$ nonnegative $\rightarrow$ signed:

Nonnegative infinite-expectation example:

Undefined signed-expectation example:

Atomic and density representations:

Exercise 2 – Tonelli and transformations

Subgraph measurability representation:

Why Tonelli applies:

Deductible identity:

Cap identity:

Limited-loss identity:

Exponential calculation:

Jensen boundary:

Exercises 3–4 – Finite conditioning

Form forced by measurability:

Coefficients forced by integral identities:

Null-atom version freedom:

Bayes normalization:

Event-conditioning RN density:

Why continuous event-ratio conditioning fails:

Representation ledger

Object	Numerator or construction	Reference	Domain	Equality basis
$d(f\mathbb{P})/d\mathbb{P}$				
$\mu_X = X_{\#}\mathbb{P}$				
$d\mu_X/d\lambda$, only if $\mu_X \ll \lambda$				
$d\mathbb{P}_A/d\mathbb{P}$				
$\mathbb{E}[Y \mid \mathcal{G}]$				

Deployment-mechanism audit

Q50 anchor – hidden transformed payment:
Q386 anchor – required nonlinear moment:
Q370 anchor – joint masses and normalization:

Final verdict

State: rupture / reconstructible / deployable
Can recover LOTUS without notes: yes / no
Can recover the tail-payment formulas without notes: yes / no
Can derive Bayes from joint masses: yes / no
First deferred proof debt:
If the last line says only “not yet familiar,” the diagnosis is incomplete. Name the missing object, hypothesis, representation, or theorem.